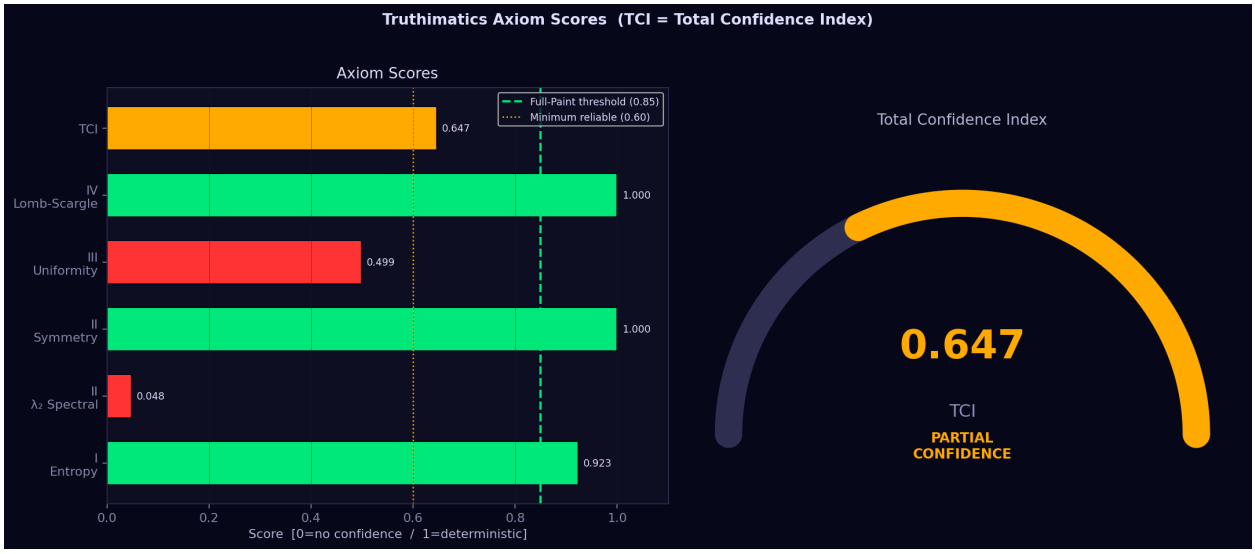
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_C7810532C4
Taxa	Deinocheirus
Clade	Theropod
PBDB Records	5
Mass (regression)	6382.3 kg (Campione & Evans 2012)
Mass (volumetric)	2485.5 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	2.924 m^3
Validity Score	1.0000 / 1.0000
TCI	0.6470 (partial)
Axiom I Entropy	0.9231
Axiom II λ■	0.1938 (symmetry 100%)
Axiom III Uniformity	0.4991
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	60.0% Measured / 40.0% Inferred
Elements	25
CoG (m)	[0.419, 0.938, 0.0]
Generated	2026-04-30 23:54 UTC

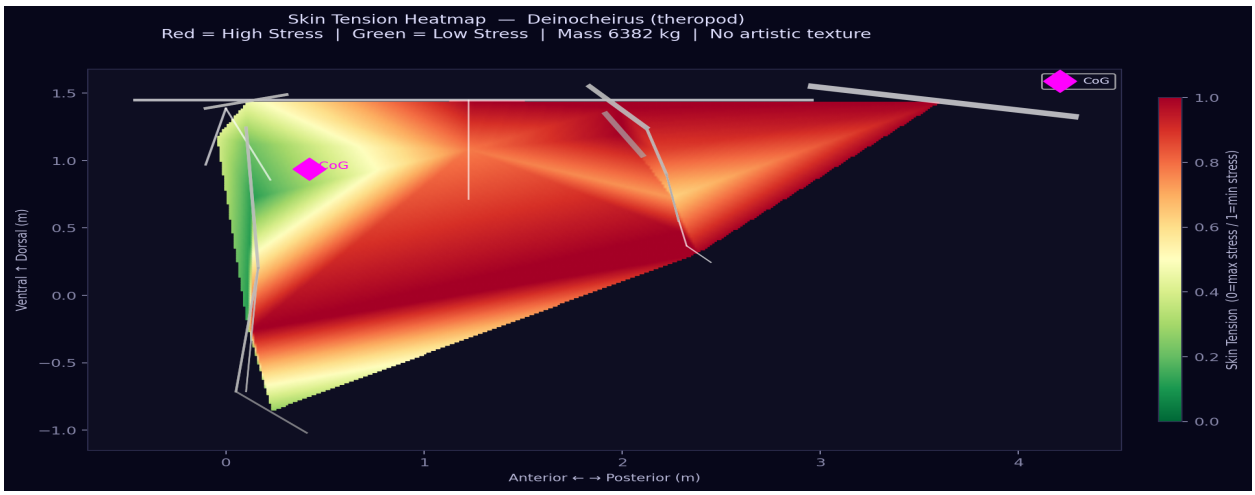
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



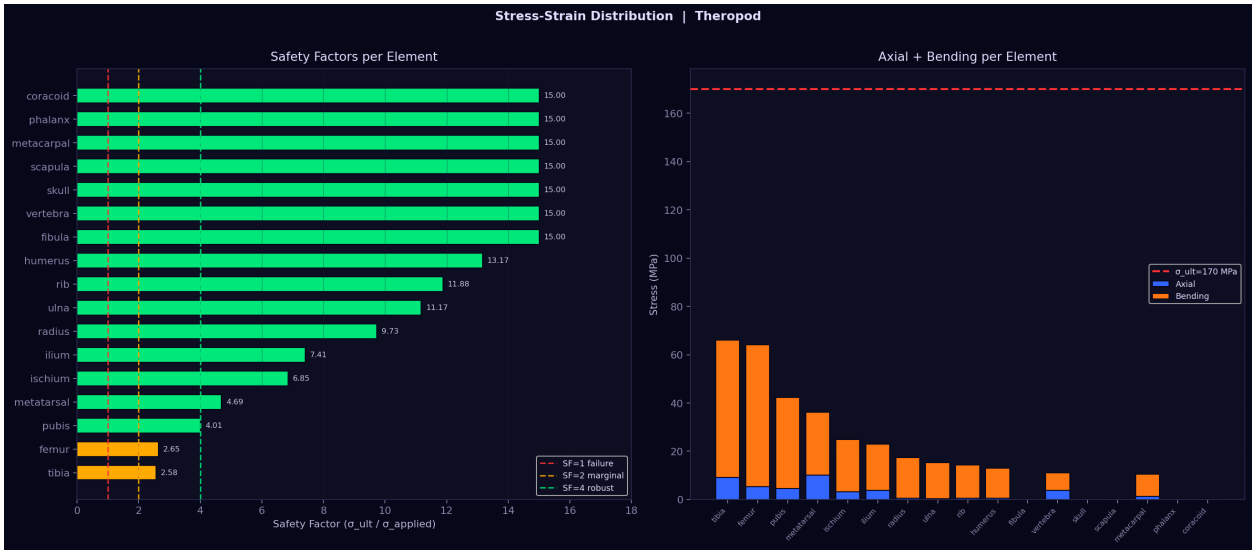
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

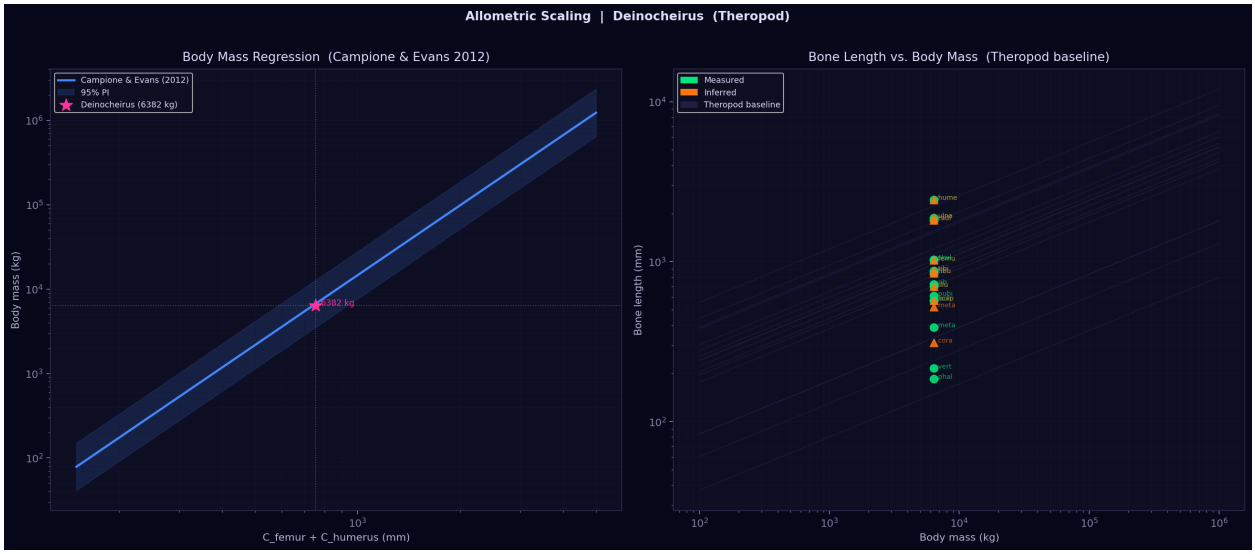


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.038	0.065	0.103	15.000	OK
femur	5.251	58.928	64.179	2.649	WARNING
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.449	12.463	12.912	13.166	OK
ilium	3.717	19.215	22.932	7.413	OK
ischium	3.232	21.577	24.809	6.852	OK
metacarpal	1.390	9.170	10.560	15.000	OK
metatarsal	10.206	26.022	36.228	4.692	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	4.496	37.881	42.377	4.012	OK
radius	0.473	17.002	17.475	9.728	OK
rib	0.451	13.861	14.312	11.878	OK
scapula	0.046	0.162	0.208	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	9.234	56.777	66.011	2.575	WARNING
ulna	0.422	14.795	15.217	11.172	OK
vertebra	3.717	7.398	11.116	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.417 m	Femur + tibia + metatarsal
Estimated Speed	16.3 km/h (4.54 m/s)	Gait: Trot | Fr=0.868
Stride Length	5.314 m	Alexander (1976) / Thulborn (1990)
Bite Force	14686.4 N (1497.1 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	1030.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3758	Circumference / length (dimensionless)
Limb Strength Index	2.6e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	8.66 m	Allometric from femur length
Est. Neck Length	1.21 m	Allometric from femur length
Est. Tail Length	3.29 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	311.9	241.3	7274.36	9080.6	Constraint	0.40	15.00
femur	1020.0	122.0	6081.07	7591.0	Measured	0.88	2.65
femur_right	1020.0	122.0	6081.07	7591.0	Symmetry	0.84	2.65
fibula	848.0	35.0	293.71	308.0	Measured	0.88	15.00
fibula_right	848.0	35.0	293.71	308.0	Symmetry	0.84	15.00
humerus	2440.0	118.0	13608.61	16987.6	Measured	0.88	13.17
humerus_right	2440.0	118.0	13608.61	16987.6	Symmetry	0.84	13.17
ilium	698.0	145.0	5878.30	7337.9	Measured	0.88	7.41
ilium_right	698.0	145.0	5878.30	7337.9	Symmetry	0.84	7.41
ischium	572.0	92.0	1939.24	2420.8	Measured	0.88	6.85
metacarpal	388.0	65.0	244.63	201.3	Measured	0.88	15.00
metatarsal	519.4	87.5	1593.26	1988.9	Constraint	0.65	4.69
phalanx	185.0	42.0	39.37	30.5	Measured	0.88	15.00
pubis	612.0	78.0	1491.42	1861.7	Measured	0.88	4.01
radius	1820.0	68.0	3370.93	4207.9	Measured	0.88	9.73
radius_right	1820.0	68.0	3370.93	4207.9	Symmetry	0.84	9.73
rib	725.0	38.0	419.34	523.5	Measured	0.88	11.88
scapula	572.0	218.0	10888.53	13592.1	Measured	0.88	15.00
scapula_right	572.0	218.0	10888.53	13592.1	Symmetry	0.84	15.00
skull	1030.0	488.0	98250.99	122646.7	Measured	0.88	15.00
tibia	878.0	92.0	2976.67	3715.8	Measured	0.88	2.58
tibia_right	878.0	92.0	2976.67	3715.8	Symmetry	0.84	2.58
ulna	1880.0	72.0	3903.76	4873.1	Measured	0.88	11.17
ulna_right	1880.0	72.0	3903.76	4873.1	Symmetry	0.84	11.17
vertebra	215.0	145.0	1810.65	2260.2	Measured	0.88	15.00